Peking University Algebra (Liang Xiao) Exe I First homework

1. 9.21 G be the group

1.1. If $x, g \in G$. prove that $|x| = |gxg^{-1}|$ Deduce that $|ab| = |ba|$. $\forall a, b \in G$.

Pf: 1) let $d = \mathrm{ord}(x)$ then $(gxg^{-1})^n = gx^ng^{-1} = e \iff x^n = e$ Thus $d = \mathrm{ord}(gxg^{-1})$.

2) $ba = a^{-1}(ab)a \Rightarrow |ba| = |ab|$ □

1.2. $n, m \in \mathbb{N}_{>2}$ Determine all homomorphisms $\phi: Z_n \to D_{2m}$.

Pf: $D_{2m} = \langle r, s \mid r^m = e, s^2 = e, srs = r^{-1} \rangle$ $\phi(Z_n) \le D_{2m}$ is the Abel-subgroup with the order $t \mid n$

There is a simple observation that $\begin{cases} sr^k s^{-1} = r^{-k} \Rightarrow sr^k = r^{-k}s \\ srs = r^{-1}, \quad rsr = s \end{cases}$

It's easy to check that every element in D_{2m} is of the form $r^k s$ or r^k

Then $\phi(Z_n)$ is commutative $\Rightarrow r^k s r^l = r^l s r^k \Rightarrow r^{k-l} = r^{l-k} \Rightarrow r^{2(k-l)} = e \Rightarrow m \mid 2(k-l)$

1°. $\phi(1) = r^k$, then $\phi(Z_n) \trianglelefteq \langle r \rangle$ where $r^{kn} = e \Rightarrow m \mid kn$

$\Rightarrow k$ is of the form $l \cdot \frac{m}{\gcd(m,n)}$ $\Rightarrow \left(1 \le l \le \frac{n}{\gcd(m,n)}\right)$

2°. $\phi(1) = r^k s$. then $\phi(2) = r^k s r^k s = 1$. $\Rightarrow 2 \mid n$. $\phi(Z_n) \cong Z_2$.

Therefore, there're two kinds of homomorphisms □

1.3. Prove that S_n is generated by (12) and (12.. n)

Pf: See 1.6.5. □

1.4. Show that an element has order 2 in S_n iff its cycle decomposition is a product of commuting 2-cycles. In S_6, how many elements have order 2?

Pf: It suffices to show "$\Rightarrow$": Suppose $a \in S_n$, $\mathrm{Ord}(a) = 2$.

Let $a = r_1 r_2 \dots r_s$ Each r_i is a t_i-cycle that doesn't intersect with others.

If $\exists t_i \ge 3$, then $a^2 = r_1^2 \dots r_s^2 \cdot r_i^2 \ne e$ (since $r_i^2 \ne e$). Contradiction!

In S_6, there're 15 many 2-cycles, By commuting, there're $C_6^2 + C_6^4 \times 3 + C_6^6 \times 3 \times 3 = 69$ many 2-elements. □

1.5. Show that D_8 and Q_8 are not isomorphic.

Pf: $Q_8 = \langle i, j, k \mid i^2 = j^2 = k^2 = ijk = -1 \rangle$. Then Q_8 has 1 element of order 2 (-1)

But D_8 has r^2, s, rs, r^2s, r^3s with order 2 □

1.6. Prove that every subgroup of a cyclic group is cyclic, and every quotient group of a cyclic group is cyclic.

Pf: ① $\mathbb{Z}$. Since every subgroup of $\mathbb{Z}$ is of the form $m\mathbb{Z}$, thus cyclic.

$\mathbb{Z}/m\mathbb{Z}$ is also cyclic

② Z_m. Since $\forall H \trianglelefteq Z_m$ $\exists a \in H$ s.t. $a \ne 0$ If $\exists b \in H$ s.t. $b \notin \langle a \rangle$ $b = aq + r$ $0 < r < a \in H$

Contradiction! Thus $H = \langle a \rangle = \mathbb{Z}_d$ $(d \mid n)$ cyclic $\quad \mathbb{Z}_n / \mathbb{Z}_d \cong \mathbb{Z}_{\frac{n}{d}}$. It also cyclic $\square$

I.7 Define an opposite group G^{op}, the underlying set is G, $a * b = ba$

prove: (1) G^{op} is a group (2) $G \cong G^{op}$

Pf (1) Obvious; (2) $\phi: G \to G^{op} \quad a \mapsto a^{-1}$ Then $\begin{cases} \phi(ab) = \phi(a)\phi(b) = b^{-1}a^{-1} \\ \phi \text{ is bijective} \end{cases}$ $\square$

I.8. (Skipped)

I.9. Consider the group $G = \mathbb{Z}_4 \times \mathbb{Z}_4$ Let $\overline{G} = G / \langle x^2 y^2 \rangle$

Show that (1) $|\overline{G}| = 8$

(2) Exhibit each element of $\overline{G}$ in the form of $\overline{x}^a \overline{y}^b$

(3) Give an explicit isomorphism $\overline{G} \cong \mathbb{Z}_4 \times \mathbb{Z}_2$

Pf: (1) $|\overline{G}| = |G| / \text{Ord}(x^2y^2) = 16/2 = 8$.

(2) $\overline{G} = \{1, \overline{y}, \overline{x}, \overline{x}\overline{y}, \overline{x}^{-1}, \overline{x}^{-1}\overline{y}, \overline{x}^2, \overline{x}^2\overline{y}\}$

(3) From (2), we have $\phi: \overline{G} \to \mathbb{Z}_4 \times \mathbb{Z}_2$

$\overline{x} \mapsto (1, 0)$

$\overline{y} \mapsto (1, 1)$.

Then it's easy to check bijection & homomorphism $\square$

I.10. $N \leq G$, $|N| < \infty$. Show that $g \in G$, $gNg^{-1} \subseteq N$ iff $gNg^{-1} = N$

Give an example where this fails for an infinite N.

Pf: Since $|N| < \infty$, we have $|gNg^{-1}| = |N|$. $\Rightarrow$ $gNg^{-1} = N$

Example: Let $G = SL_2(\mathbb{Z})$ N be the group $:= \left\{ \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \middle| t \in \mathbb{Z} \right\}$

By commuting, $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$, $gNg^{-1} \subseteq N \iff c = 0$.

And we have $gNg^{-1} = \left\{ \begin{pmatrix} 1 & ta^2 \\ 0 & 1 \end{pmatrix} \middle| t \in \mathbb{Z} \right\} \subsetneq N$ Since $\exists\, b \in \mathbb{Z}$ s.t. $(a^2, b) = 1$ [

I.11. Let $D_{2n} = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle$, $k \mid n$. Show that

(i) $\langle r^k \rangle \trianglelefteq D_{2n}$

(ii) $D_{2n} / \langle r^k \rangle \cong D_{2k}$

Pf: (i). $\forall g \in D_{2n}$. say $\begin{cases} r^l s \\ r^l \end{cases}$, we have $r^l s r^k r^l s = r^l r^{-k-l} = r^{-k}$ with the same order of r^k

we have $r^l r^k r^{-l} = r^k$

$\Rightarrow g \langle r^k \rangle g^{-1} = \langle r^k \rangle$.

(ii). $\overline{r}^k = 1$, $\overline{s}^2 = 1$. $\Rightarrow$ $\langle \overline{r}, \overline{s} \mid \overline{r}^k = \overline{s}^2 = 1, \overline{rs} = \overline{s}\overline{r}^{-1} \rangle \leq D_{2n} / \langle r^k \rangle$

Also $|D_{2n} / \langle r^k \rangle| = 2n / \frac{n}{k} = 2k$ Thus $D_{2n} / \langle r^k \rangle \cong D_{2k}$ $\square$

$\leq mn$

$xH, xK) = (yH, yK) \Rightarrow xy^{-1} \in H \cap K \Rightarrow x(H \cap K) = y(H \cap K) \Rightarrow$ injective

Thus $|G / H \cap K| \leq mn$

On the other hand

I.1.12. Any subgroup H of index 2 of G is normal

Pf: $[G:H]=2$, If $\exists g\in G$ s.t. $gHg^{-1}\neq H$

Then $gH\neq Hg$, $g\notin H$ Moreover, $G=H\sqcup gH=H\sqcup Hg \Rightarrow gH=Hg$ Contradiction! □

I.1.13. Let $H_1,H_2\leq G$, and let $N\trianglelefteq G$ contained in $H_1\cap H_2$. Show that $(H_1/N)\cap(H_2/N)\cong(H_1\cap H_2)/N$

Here the intersection takes place in G/N

Pf: Define $\pi: H_1\cap H_2\longrightarrow (H_1/N)\cap(H_2/N)$

$$x\longmapsto xN\in(H_1/N)\cap(H_2/N)$$

Then ① $\forall x,y\in H_1\cap H_2$, $\pi(xy)=xgN=xN\,gN$ $(N\trianglelefteq G)$.

② $\forall xN\in(H_1/N)\cap(H_2/N)$, $\pi(x)=xN$ ③ $\ker\pi=N$.

$\Rightarrow$ $\dfrac{H_1\cap H_2}{N}\cong(H_1/N)\cap(H_2/N)$ □

I.1.14. $|G|<\infty$. Let $H,K\leq G$ and let $a\in G$. $HaK:=\{hak\mid h\in H, a\in K\}$. Prove that $|HaK|=|H|\cdot|K:a^{-1}Ha\cap K|$

Pf: (skip) See Homework II.11.

I.1.15. Let $H,K,N\leq G$ such that $H\leq K$, $H\cap N=K\cap N$, and $HN=KN$ Show that $H=K$.

Pf: Lemma: There is a bijection (set) between $H/H\cap N$ and HN/N

$$\varphi: H/H\cap N\longrightarrow HN/N \qquad h(H\cap N)\longmapsto h(H\cap N)N=hN$$

① It's well-defined, $h_1(H\cap N)=h_2(H\cap N)\Rightarrow h_1h_2^{-1}\leq H\cap N\leq N\Rightarrow h_1N=h_2N$.

② Surjective, $\forall hnN\in HN/N$ namely hN, there is $\varphi(h(H\cap N))=hN$.

③ injective, $h_1N=h_2N\Rightarrow h_1h_2^{-1}\in H\cap N\Rightarrow h_1(H\cap N)=h_2(H\cap N)$

Back to the proof: We have $H/(H\cap N)\overset{\varphi}{\longleftrightarrow} HN/N=KN/N\overset{\phi}{\longleftrightarrow} K/(K\cap N)$.

$\Rightarrow$ $H/(H\cap N)$ and $K/(K\cap N)$ are isomorphic (set).

$\pi_1: H\longrightarrow H/(H\cap N)$; $\pi_2: K\longrightarrow K/K\cap N$

Def: $\psi: H\to K$. $\psi:=\pi_2^{-1}\circ\phi\circ\varphi\circ\pi_1$ It's easy to check that it's an isomorphism (Set) □.

$$\begin{array}{ccc} H & \xrightarrow{\psi} & K \\ \downarrow{\scriptstyle\pi_1} & & \downarrow{\scriptstyle\pi_2} \\ H/H\cap N & \xrightarrow{\phi\circ\varphi} & K/K\cap N \\ \| & & \| \\ H/S & & K/S. \end{array}$$

II.1.16. Suppose that $H,K\leq G$ with finite index where $[G:H]=m$, $[G:K]=n$.

Show that (1) $\operatorname{lcm}(m,n)\leq[G:H\cap K]\leq mn$.

(2) If $\gcd(m,n)=1$, then $[G:H\cap K]=[G:H]\cdot[G:K]$.

Since $[G:H\cap K]=[G:H]\cdot[H:H\cap K]$

$\Rightarrow m\mid[G:H\cap K]$

$\Rightarrow \operatorname{lcm}(m,n)\leq[G:H\cap K]\leq mn$

Pf: Only (1): Consider the map $\varphi: G/H\cap K\longrightarrow G/H\times G/K$.

$$x(H\cap K)\longmapsto (xH,xK)$$

If $x(H\cap K)=y(H\cap K)\Rightarrow xy^{-1}\in H\cap K\Rightarrow xH=yH$, $xK=yK\Rightarrow$ well-defined

If $(xH,xK)=(yH,yK)\Rightarrow xy^{-1}\in H\cap K\Rightarrow x(H\cap K)=y(H\cap K)\Rightarrow$ injective.

Thus $|G/H\cap K|\leq mn$

On the other hand

I.1.17. Suppose that $H, K \trianglelefteq G$. and that G/H & G/K are both solvable.
Prove that $G/(H \cap K)$ is solvable

Pf: $\{1\} = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_{s_1} = G/H$, G_i/G_{i-1} Abelian

$\{1\} = G_0' \trianglelefteq G_1' \trianglelefteq \cdots \trianglelefteq G_{s_2}' = G/K$, G_i'/G_{i-1}' Abelian

Thm

2. More exe.

I.2.1. List all subgroups of D_{2n} and decide which subgroup is normal.

Pf: Since every element of D_{2n} is of the form $r^k s$ or r^k,

we have $\langle r^k \rangle \cong Z_{\frac{n}{k}}$. $k|n$. , $\langle s \rangle \cong Z_2$. , $\langle r^k s, r^l \rangle (l|n) = \{r^l, \dots, 1, r^{k+l}, \dots, r^k s$

$\cong D_{2\frac{n}{l}}$

$\langle r^k s, s \rangle = \langle r^k, s \rangle \cong D_{2l}$. (for some $l|n$.)

Therefore, when $2|n$. all subgroups (under isomorphism) are Z_d $(d|n)$, D_{2d} $(d|n)$

when $2 \nmid n$, all subgroups are Z_d $(d|n)$, Z_2, D_{2d} $(d|n)$.

Clearly, we have that $Z_d \trianglelefteq D_{2n}$ $\forall d|n$ $\square$.

I.2.2. Let $H \leq G$, $|G| < \infty$. Show that $\exists$ $g_1, \dots, g_n$ simultaneously representing left and right cosets of H in G, that is $G/H = \{c_1 H, \dots, c_n H\}$. $H \backslash G = \{Hc_1, \dots, Hc_n\}$.